# Kerf Guided Glulam

## A novel way of creating curved glulam beams

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*This paper proposes a novel way of producing curved glulam timber elements where the formwork is integrated into a glulam beam. The method proposed accomplishes this by placing kerf cuts on a timber profile that gets bent and then encased in a wood laminate, forming the glulam beam. The kerf placement allows the beam to be asymmetrically curved. The optimal placement for the kerf cuts is found by feeding an initial goal curve to a form-finding definition that subdivides it and places markers where cuts need to be made while manipulating the beam geometry, ensuring that it matches the initial input curve. The benefit of this method is that it is not reliant on large-scale glulam setups but can be fabricated with basic wood workshop tools in conjunction with a 5-axis CNC mill. The simplified production process enables smaller manufacturers and designers to produce dynamic wooden structures while saving on materials and labour that would have gone into producing formwork that eventually gets discarded.*

**Keywords:** *Digital Wood Workflows, Kerfs, Glulam, Parametric Design, Digital Fabrication, CNC, Design Democratisation.*

## INTRODUCTION

Recent innovations in glulam fabrication have centred around further utilisations of diverse wood species and grades in conjunction with implementing newer glue types and inserts such as synthetic fibres (Silvans 2020). Micro-cutting or kerfing has had much research on sheet goods resulting in non-deformable materials being transformed into double-curved elements (Capone 2018). Regarding beams, the research has been focused on the methodology of kerf cutting, focusing on variable curvature and out-of-plane deformations (Naboni 2021), with little to no research being done into the combination of glulam and kerf cutting.

When it comes to the structure of laminate elements, the relatively mature production method has stayed the same, i.e., similar layers of laminate that get glued together. This method for producing glulam elements requires highly specialised dynamic clamping systems (Menges et al., 2021) or specially cut jigs for curved glulam elements (How, Etal., 2016).

The production methods limit the fabrication to either wood workshops specialising in glulam production or necessitate the creation of single-use jigs, wasting labour, funds, and material. By using the innate characteristics of wood, i.e., its relatively low stiffness in contrast to its high strength (Menges et al. 2021), in conjunction with its anisotropic qualities, where material can be removed without entirely compromising its structural integrity (Menges 2010), wood can be made to bend in a controlled fashion that can be programmed into it by placing cuts at strategic points. Varying the placement of these cuts,

called kerfs, allows the user to convert asymmetrical curves into polylines by finding points where the polylines can replace the curves with minimal deviation. The benefits of integrating the formwork into the beam via the kerf cut lengths of wood are that the material and labour that would go into producing traditional glulam formwork get used in the final construction, thereby eliminating excess use of materials.

## Project constraints

The impetus for the development of the system was an exhibition project where a basket-like structure needed to be made of beams with varying curvatures. While the system itself was not used in the final construction due to the fabricator choosing to go with a more established production method, it allowed for the construction of prototypes. The exhibition project supplied the goal geometry for the said prototypes.

The material chosen for the prototypes shown in this paper is construction-grade pine (C18). This choice was made because it was readily available and had beneficial properties such as ease of workability, good elastic performance and small knot size. When choosing materials for the kerf core, the least number of knots is desirable, while the laminate strips themselves can be of poorer quality due to them bracing against each other.

The elements produced have a cross-section of 60 by 120 millimetres and a length ranging from 1.8 to 2 meters.

The kerf-core is set at one-third of the beam's final dimensions, leaving the other two-thirds to be made up of the laminate layers; this is done to leave enough material for milling and post-processing. The core thickness was set at 45 millimetres for the prototypes shown here.

The kerf-cores were milled and sawed on a five-axis CNC with a work bed size of 2.5 by 1.5 meters using both endmills and a sawblade.

## Digital Process

The digital process starts by defining the centre curve of the intended beam. This can be any curve that bends in one direction. Then, the beam thickness, sawblade width, kerf depth and desired overlap degree are defined. The beam thickness and blade width are used to calculate the angle needed to close the kerfs. The input curve is then offset half the width of the beam thickness creating the exterior boundary of the beam.

With the exterior boundary curve defined and the kerf angle set, these are used to determine goals for an iterative Kangaroo solver built into the parametric Rhino/Grasshopper environment. This goal setting starts by determining the angle between the tangent vectors of the curve's endpoints (Figure 1).

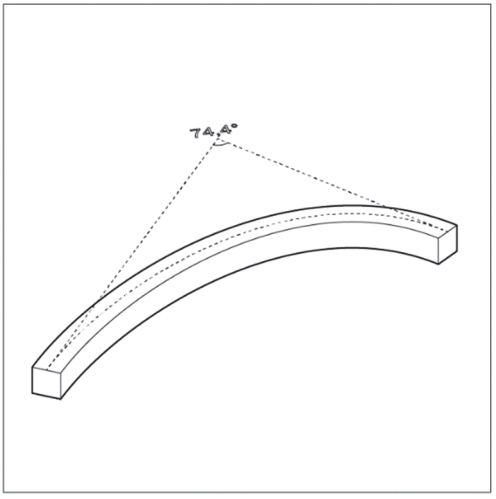



Figure 1
The intersection of the two endpoint tangent vectors together with the inverse kerf angle defines the number of kerf cuts needed to produce the wanted bend in the beam.

This angle is then divided by the inverse angle of the kerf angle resulting in the number of times the curve needs to be divided to achieve the desired polyline of the kerf beam. The number generated by the division has a floating-point value rounded to the nearest whole number; this introduces some imprecision in the angles, which will be addressed later.

The segmented curve is now used to generate a polyline, of which the control points are used to create goals to keep the polyline on the curve. The

endpoints of the polyline are used as anchors, while the goal angle between the polyline segments is set to the inverse kerf angle. When the goals have been set for the Kangaroo definition, optimisation can be initiated. The fitness of the resulting polyline can be measured by inspecting the inverse kerf angle compared to the arithmetic mean of the angles generated through the Kangaroo optimisation; typically, the discrepancy between the goal and the result is within 0.2 degrees. This can be further decreased by running the optimisation for longer but suffers from diminishing returns.

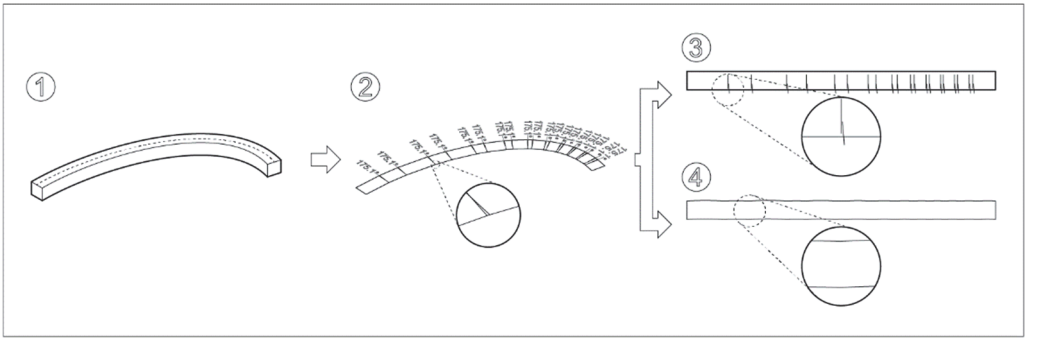



Figure 2
Computational workflow
1. The beam center curve gets defined
2. Cuts get spaced along the beam relief cuts are generated with geometry overlap
3. Cutting geometry for kerfs and relief angle gets generated
4. Scalloping boundaries generated

When the curve has been optimised, its control points are then used to define the placement of the kerf cuts. The control points are then extended into a line that defines the kerf cuts needed in the beam.

The optimised curve is used for simulating how the kerfs close. This is where the overlap degree is used; when it was defined, it got added to the kerf angle forcing an overlap of the kerf joints the overlap angle ensures that the kerf bends not only meet at a point but also overlap. The reason for this is twofold, the relief generated by the overlap ensures that the meeting between the beam segments is a solid end-to-end connection strengthening the bonding of the elements rather than a point that could get deformed when the beam is tensioned and glued; the second reason is that the relief cuts are all at slightly different angles ensuring that the discrepancies generated when the curve was optimised get corrected.

The original offset curve gets split by the optimised control points, and the resulting curve segments are oriented to create a vertical series forming the outer perimeter of the milled beam. The segments then get offset by the beam thickness, defining the milled beam's inner perimeter (Figure 2). These curve segments form scallop shapes in the beam that line up when it gets clamped together, forming a curve matching the input curve rather than a segmented line shape. The beam being a curve rather than a segmented line also prevents it from delaminating when the strips of the glue laminate are added to the outside (Figure 3, 4).

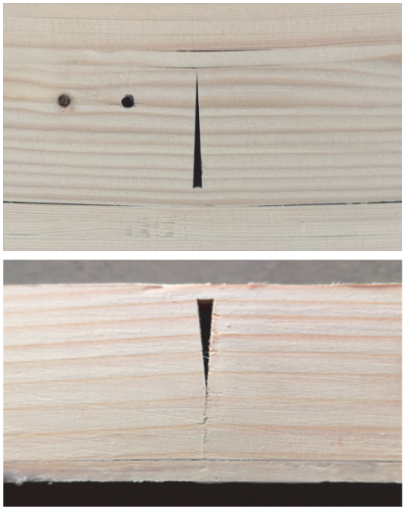


Figure 3 and 4
Example of the delamination that happens when the beam is angled in contrast to an example of a curved laminated beam.

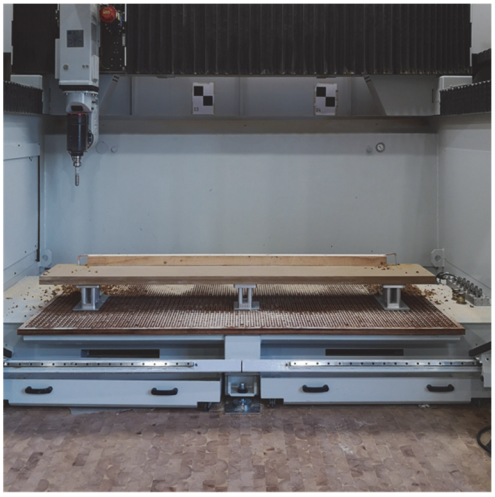


Figure 5
Clamping setup using screw clamps at the ends and double-sided tape underneath the material blank.



Figure 6
Kerf with relief angle cut guiding the beam into the correct curvature when tensioned.

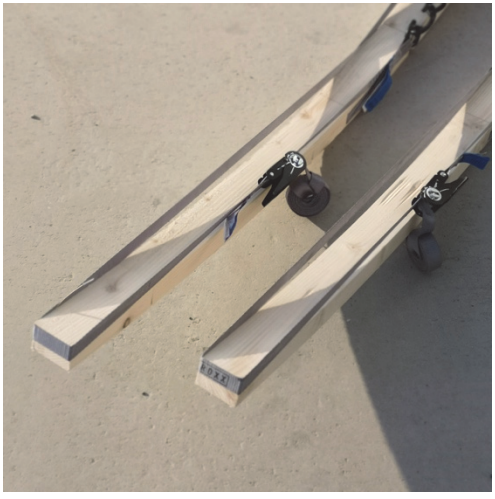


Figure 7
Examples of a double strapped beams where the tension gets equally distributed through the kerf cuts.

Lastly, the kerf cuts get oriented to the scalloping cuts forming the cutting instructions for the CNC operations. Cutting instructions can be exported as curves and lines or solid three-dimensional geometry, depending on the manufacturer's preferences.

## Pre-production, milling and assembly

The pre-production starts with planning the rough construction lumber to the desired thickness (Z-height) and then cutting the beam elements to the required width (Y-width). The Grasshopper script supplies the minimum beam thickness, as 5 mm safety is added to the width and 2.5 mm to the thickness. The blank is then cut to the beam's required length (X-length). All the values needed for the pre-production can be accessed in the Grasshopper definition. The offcuts can be then turned into strips for the glulam layers. The resulting blank is mounted into the 5-axis CNC machine with simple methods such as a waste plate, clamps and double-sided tape (Figure 5) or more advanced tools like adaptable pneumatic clamps.

The kerf cut beam is produced in two passes, one milling and one sawing operation. The scallops get milled first while the sawing operations are done from the side; this enables both procedures to be carried out in one setup eliminating errors and alignment issues that might appear if the beam was moved and reoriented between operations. During the milling operation, the end cuts are skipped to have areas to clamp the beam down. This first pass produces the scallops. The second pass is made with the saw cutting the kerfs and the relief angle of the joints (Figure 6).

When the milling passes are finished, the beam can be taken out of the mill and tensioned with band clamps. This relieves excess pressure on the kerf joints when the beam gets tensioned from the ends (Figure 7). The beam is adequately tensioned when the kerfs and relief cuts are closed. Special care needs to be made that the strap part of the tensioning clamps doesn't overlap the relief cut, as this will prevent the joints from closing correctly.

When the beam has been tensioned, the lamination process can be started by locking the kerfs together and adding the desired number of layers to each side of the kerf-beam, achieving its nominal thickness. The best practice is to clamp from the middle out, ensuring good bonding between the different layers of the glulam and ensuring that the layers lie flat and in-plane with the top and bottom surfaces of the beam.

The beam's geometry can be verified using digital means such as photos, lidar scanning, or manual means like physical printouts. After the beam has been approved, it can be post-processed on the CNC mill or by hand.

## Results and findings

The method is subject to a series of limitations due to the material characteristics of wood and how the data of the beams get processed. In addition, some aspects of the system's rigidity have not yet been tested.

Knot positions are essential when choosing the base material for the kerf beam. Structural lumber of at least strength grade C18 or higher is recommended. The overlap of knots and the kerfs will cause structural failure when the beam is tensioned. The side of the lumber with the least number of knots is placed so that it faces the opposite side of the kerf cuts to maintain the structural integrity of the beam.

The curves implemented into the Grasshopper definition can only be curved in one direction, for example, from left to right. The optimisation script does not detect the point on the curve where it changes direction, throwing the results off, for instance, in the case of an S-shaped beam.

The kerfing of the wood leads to many instances of exposed end grain where the wood gets more receptible to water ingress; therefore, in their current form, the beams created with this method should not be used in situations where water can seep into the joints leading to premature decomposition of the beams.

The strength and stiffness of the kerf cut beam have not yet been compared to a traditionally produced glulam beam (Figure 8).



Figure 8
A single layer kerf guided glulam beam

## Conclusion

The proposed method is not meant to replace the established industry practices for creating curved glulam elements; instead, it is a new workflow that allows laypersons to design dynamic wooden structures that would typically be hard to fabricate without using highly specialised equipment. The method proposed here is still in its infancy, and there is ample room for improvement. The future perspectives for developing the system could include kerfing that generates out-of-plane deformations allowing the creation of doubly-curved glulam blanks and the implementation of "exclusion zones" where the kerf cuts get pushed away, enabling more detailed programming of the beams where material can be left for joints to be made. Most traditional joinery techniques will work when it comes to joining the beams. Still, developing a more native joinery system for this particular beam type would be relevant, where the kerf cut core is avoided, and the laminates are used as the contact points. As of now, the beams produced through the research and testing of this paper

are suited to medium-sized structures such as pavilions, where a high degree of geometric customisation is required. The use of the beams as a structural element is dependent on the result of future stress testing comparing them to more traditional glulam elements.

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